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Restoring Nature

Incentivising and delivering on sustainable and equitable ecosystem restoration

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A biodiversity conservation protocol: an integrated community-based model

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Please indicate your preferred method of presentation: Oral

Problem: This is an integrated community-based proposal to restore the natural resources for mutual benefits while generating the next generation through practical awareness of their importance and why conserve them via available scientific and technological interventions. Forest fires is one of the major threats to biodiversity declination including plants and their eventual extinction. It has major detrimental impacts on the environment especially in rural parts of the developing world societies whose people are to some extent largely disadvantaged economically both in education as well as communication regarding the importance of natural products resources. Some of the natural resources occurring in various geographical parts of the world are unique to that habitat only.

Evidence: This model is an attempt to create awareness of the importance and the need to conserve indigenous medicinal plants in Kairini hill in Tharaka Nithi County of Kenya, that is anchored on the local community in collaboration with the local and national government agencies, the county and national government, research scientists, and institutions of higher learning thereby restoring nature in line with sustainable development goals 13 and 15 on climate and life on land respectively a together with national policies on environment and climate change. The proposed protocol is based on scientific findings that this geographical region is rich in indigenous medicinal plants with antimicrobial potentiality (Mutembei *et al*., 2018) and worthy further scientific investigation and conservation for the next generation due to the threat of perennial annual burning and charcoal burning by the local community. Our current on-going investigation on the potentiality of these plants in addressing human fertility and safety reveals a huge an untapped potential that is yet to be explored scientifically given that most of this knowledge is passed on orally from one generation to the next. It has been observed that there exist a gap in awareness between the local people and the government efforts to restore the plants. Due to charcoal burning as a means of livelihood, the local people view the government agencies such The Kenya Forestry as an enemy and thus any forum for a sustainable plant conservation interaction is a serious challenge. It has been reported that indigenous knowledge, values, and cultural practices by people in various societies on earth is an opportunity to address various challenges including biodiversity conservation (Brondízio *et al*., 2021) and thus it is important to integrate them in any undertaking be it locally or otherwise. For example, Nyambene hills in Meru County of Kenya is as a result of integrating the Igembe and Tigania ethnic communities to restore its flora. Such a community based organization to conserve environment lacks in Tharaka Nithi County.

Solution: The model aims to extent to the whole county integrating the particular ethnic communities as an attempt to meet the national goals on mitigating climatic changes and their associated detrimental effects such as drought. The protocol proposes to engage youths and pupils at elementary level of education by planned regular planting of trees especially the indigenous ones that thrives well in these habitats through the already ongoing tree planting exercise by the Kenya Defense Forces under the ministry of defense. The model also proposes a section for setting up a botanical garden within the hill from which, through the county government coordination can be a source of revenue collected by visitors as a self-sustainability mechanism in the future. This borrows an already established botanical garden at Sino Africa Joint Research Centre hosted in Jomo Kenyatta University of Agrciture and Technology, JKUAT commonly known as SAJOREC-JKUAT here in Kenya. For years, the local community have been burning the grass-rich hill just before October-December short rains (undocumented observation from the local forestry department). Even though unaware of the scientific principle behind this old-age practice (to break seed dormancy of grass seeds) the people know that the green grass sprouts for their livestock. This has been a big challenge on efforts by government to curb this practice. In collaboration with the engineering department, some JKUAT mechatronics engineering students, another key collaborating institution of higher learning in this work are developing, as part of their final academic projects, drones mounted with fire detectors monitored from a control station as an attempt to address this challenge. Chuka University, the local public university bordering this locality is proposed to play a critical role in realizing this aim especially mentorship of the surrounding primary and secondary schools and thus raising a new generation aware of the usefulness of flora and fauna thereby protecting them.

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